

MedIntel AI – Hospital Management Analytics

Project Description

MedIntel AI is an end-to-end Hospital Management Analytics project developed to analyze hospital operations and support data-driven decision-making. The project integrates PostgreSQL, SQL, Python, and Power BI to simulate a real-world hospital management system and generate business insights through interactive dashboards.

Objectives

- Design a relational hospital database.
- Generate realistic healthcare data using Python.
- Perform SQL-based business analysis.
- Build interactive Power BI dashboards.
- Provide actionable insights for hospital management.

Technology Stack

PostgreSQL
SQL
Python
Power BI
Excel

Database Modules

Patients, Doctors, Departments, Appointments, Admissions, Billing, Payments, Pharmacy, Emergency, Insurance Claims

Dashboards

Executive Dashboard
Patient Analytics
Doctor Performance
Finance Dashboard
Emergency Dashboard
Pharmacy Dashboard
Insurance Dashboard

Key Features

End-to-end analytics solution
Interactive Power BI dashboards
SQL analytics queries
Python data generation
PostgreSQL database design
KPI cards, slicers and page navigation

Skills Demonstrated

Database Design, SQL, Python, ETL, Power BI, Data Visualization, Business Intelligence

Future Enhancements

Streamlit Web Application
Machine Learning Predictions
Real-Time Data Integration

Author

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